Mathematics Extended Essay on the topic of Erdős' Probabilistic method

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Contents

How can we use Erdős' Probabilistic method to explore clique sizes in Scale-Free Networks without requiring constructive methods?

1 Introduction

- A brief introduction - My motivation to explore this topic

To briefly help the reader to understand this essay better, below are a few definitions which might come in handy throughout the next few pages: An Objects is a

To understand the rest of the essay, it is important to clarify the concept of combinations. Imagine a simple cake in which all ingredients are mixed independent of the order. In mathematical langFurthermore, the following notation will occur commonly.

2 Scale Free Networks and Graph Theory

- Describing the concept of graphs - Defining terms - Applying the idea to scale free networks

One predominant branch of Mathematics which is often ignored is called combinatorics. It focuses on the idea of counting, which is applied in order to find a result but can also define a final answer. I will talk more about that concept using an example below. Part of combinatorics is the concept of graphs which is related to graph theory which explores the relationship between objects. These should not be confused with a graph on a cartesian plane. These graphs allow the mathematical analysis of networks and connections, like streets, mobile networks, or social media networks. From now on the use of the word graph will refer to the mathematical graph which makes part of discrete mathematics, i.e. a network of nodes connected by edges.

2.1 Definitions

For clarity, I will define the following terms, as they are different around the world:

- A <u>node</u> is the same as a vertex and is the undivisble point at which one or more edges meet.
- An <u>edge</u> is the connection between nodes.
- The <u>degree</u> of a node is the number of connections, i.e. edges, it has to another node.

I am only making use of undirected graphs, meaning that the edges don't represent a direction in the connection between nodes (as would be required when representing one-way streets for example)

2.2 Scale Free Networks

This extended essay will investigate scale free networks which are representative of many real-world networks because of their "Hubs". A scale free network gives a power a relation between the degree of a node and the probability of its occurence. In other words,

$$P_{(deg)}(k) \propto k^{-\gamma}$$

where γ is some exponent and $P_{(deg)}(k)$ the probability of a node with degree k.

To generate scale free networks, different methods exist. The greater category of the most commonly used model is called *configuration model* because it allows the degree of every node to be pre-defined. The one this extended essay focuses on is the Chung-Lu Configuration Model becauser it allows better analysis because of its dynamic charactersitics meaning it is .

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3 The Probabilistic Method

- Describing Probabilistic Notation - Applying using simple a simple Ramsey number example (R(3,3) = 6)

As we discussed above, graph theory is computationally difficult to explore. When trying to generate every possible combination of worls

4 Predicting clique sizes in Scale Free Networks

- Setting out the problem: we want to find specific clique sizes in a scale free network with 4 hubs - Using the Chung-Lu Random Graph Model, we will describe how to statitcally generate a graph, with different probabilities of nodes and edges occuring. - Using the first moment we will explain the concept of

5 Building Scale Free Networks

- Using the help of AI to code a python script, I aim to generate a statistically meaningful number of Scale-Free Networks. - By analysing the sizes of clique's present, I should be able to find similar results to my theoretical/probabilitic approach

6 Comparing Results

- Comparing results and making sure theory aligns with empirical data

7 Conclusion

- Conclude on the uses of my results